GENESIS: GENe Evaluation System with Integrated Security - A Federated Learning Framework for Detecting Gene Variant Pathogenicity

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BONAFIDE CERTIFICATE

Certified that this report for the project titled "**GENESIS: GENe Evaluation System with Integrated Security - A Federated Learning Framework for Detecting Gene Variant Pathogenicity**" is a bonafide part of the final year project carried out by **Sobhan Bose, Satirtha Goshal, Soumya Ghosh, Soumik Bag** as partial fulfillment for the degree of Bachelor of Technology in Information Technology, Maulana Abul Kalam Azad University of Technology, Kolkata, West Bengal, under my supervision.

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Abstract

The clinical interpretation of genetic variants remains a significant bottleneck in the diagnosis of rare diseases. Although machine learning models have shown great promise in predicting variant pathogenicity, their performance is fundamentally limited by access to large, diverse datasets, as strict data privacy regulations create data silos that prevent the aggregation of genomic data required for building accurate and generalizable models. In this article, we present GENESIS (GENe Evaluation System with Integrated Security), a full-stack, privacy-preserving framework for predicting the pathogenicity of Single Nucleotide Variants (SNVs) using Federated Learning (FL), offering a scalable solution to the conflict between data-driven medical research and patient privacy. GENESIS demonstrates that federated models can retain centralized-level performance—achieving an ROC-AUC of 99.54%, closely matching the non-federated baseline—while ensuring that raw genomic data never leaves local institutions. Moreover, the use of advanced aggregation strategies yields substantial gains under heterogeneous, real-world data distributions, with FedAdam improving orthogonal-set accuracy by over 20% compared to FedAvg and FedProx. By validating that high-performance pathogenicity prediction is achievable in a fully privacy-preserving, decentralized setting, this work provides a foundational blueprint for secure, collaborative genomic modeling aimed at accelerating diagnostic timelines and improving outcomes for patients with rare genetic diseases.

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Chapter – 1

Introduction

Rare diseases, although individually affecting only a small fraction of the population, collectively constitute a significant global public health challenge. Current estimates suggest that between 4% and 6% of the world's population is affected by one of the thousands of identified rare diseases, resulting in hundreds of millions of patients worldwide [1]. These disorders are highly heterogeneous in terms of their clinical manifestations, disease progression and underlying biological mechanisms. However, despite this diversity, approximately 80% of rare diseases are believed to have a genetic origin, with nearly 70% presenting during childhood [1]. Early and accurate diagnosis is therefore critical, as it enables timely clinical intervention, informed treatment planning, appropriate genetic counselling and improved patient management. Unfortunately, due to the rarity of individual conditions and the complexity of their genetic basis, patients often experience prolonged diagnostic journeys spanning several years before receiving a definitive molecular diagnosis.

The rapid advancement of next-generation sequencing technologies, particularly Whole Genome Sequencing (WGS), has fundamentally transformed the field of clinical genomics. WGS has emerged as a first-line diagnostic tool for many rare genetic disorders because of its ability to comprehensively interrogate both coding and non-coding regions of the human genome in a single experiment. Unlike targeted sequencing approaches, WGS provides an unbiased assessment of virtually all genetic variation present within an individual, thereby substantially increasing the likelihood of identifying disease-causing variants. However, the enormous volume of data generated by modern sequencing technologies has introduced a new computational challenge. A typical human genome differs from the reference genome at approximately 4 to 5 million positions, while even an exome contains nearly 20,000 genetic variants [2]. Among this vast collection of variants, only one or a very small number may be responsible for the patient's clinical phenotype. Consequently, distinguishing pathogenic variants from the overwhelming majority of benign polymorphisms has become one of the central challenges in genomic medicine.

Variant interpretation has therefore evolved into an indispensable component of genomic diagnostics. Experimental validation of every observed variant is impractical due to the immense scale of genomic data, motivating the development of computational methods capable of prioritizing variants based on their likelihood of pathogenicity. Over the past decade, numerous machine learning and statistical prediction models have been proposed to automate this process by integrating diverse sources of biological evidence, including evolutionary conservation, functional genomic annotations, protein structural information and population allele frequencies. These computational predictors enable researchers and clinicians to rapidly filter millions of candidate variants into a much smaller subset that warrants further clinical investigation.

The effectiveness of these predictive models depends heavily on the availability of large, accurately curated datasets containing clinically annotated variants. Among these resources, the ClinVar database [3] has become one of the most important repositories for variant interpretation. ClinVar is a publicly accessible archive that contains more than 2.6 million submissions describing genetic variants and their associated clinical significance, contributed by clinical laboratories, research institutions and expert review panels worldwide. The continual growth of ClinVar has enabled increasingly sophisticated machine learning models to be trained using diverse and clinically validated examples of pathogenic and benign variants.

Leveraging these large-scale annotated datasets, recent computational approaches have achieved remarkable improvements in pathogenicity prediction. Models such as MAGPIE (Multimodal Annotation Generated Pathogenic Impact Evaluator) [4] integrate a rich collection of multimodal genomic annotations with advanced machine learning techniques to provide highly accurate variant classification. Similarly, DANN [5] employs deep learning to estimate the functional consequences of genomic variants by learning complex nonlinear relationships between genomic features. These methods represent the current state of the art in computational variant interpretation and demonstrate the substantial benefits of combining comprehensive feature engineering with modern machine learning algorithms. Nevertheless, despite their impressive predictive performance, these centralized learning approaches rely upon a fundamental assumption that is becoming increasingly difficult to satisfy in real-world healthcare environments: that genomic data from multiple institutions can be collected and aggregated into a centralized repository for model development.

In practice, this assumption conflicts directly with the legal, ethical and regulatory frameworks governing genomic data sharing. Genomic information represents one of the most sensitive forms of personal data because it contains uniquely identifiable information about an individual and potentially their biological relatives. Consequently, numerous national and international regulations have been established to safeguard patient privacy and restrict the transfer of sensitive health information. Prominent examples include the General Data Protection Regulation (GDPR) in Europe and the Health Insurance Portability and Accountability Act (HIPAA) in the United States. These regulatory frameworks impose stringent requirements on how genomic data may be collected, stored, processed and shared across institutions. While these protections are essential for preserving patient confidentiality and maintaining public trust, they also create substantial barriers to collaborative machine learning. As a result, valuable genomic datasets remain isolated within hospitals, research laboratories and sequencing centers, forming institutional data silos that prevent the construction of comprehensive centralized datasets.

The fragmentation of genomic data has important consequences for the development of machine learning models. Models trained on data collected from a limited number of institutions may fail to adequately capture the genetic diversity present across different populations, sequencing technologies and clinical practices. Such models may therefore exhibit reduced generalizability when deployed in new clinical settings and may inadvertently introduce biases against underrepresented populations. Furthermore, many clinically important variants, particularly rare pathogenic variants and Variants of Uncertain Significance (VUS), occur so infrequently that meaningful statistical learning requires data from multiple institutions. The inability to securely combine these distributed datasets therefore limits both the predictive performance and the clinical applicability of existing computational pathogenicity predictors.

To overcome these limitations, an alternative machine learning paradigm is required that enables collaborative model development without requiring the exchange of sensitive patient data. Federated Learning (FL) [6] has emerged as a promising solution to this challenge. Instead of transferring raw data to a central server, FL distributes a shared model to multiple participating institutions, where the model is trained locally using each institution's private dataset. Only the learned model parameters or updates are transmitted to a coordinating server, which aggregates the contributions from all participants to produce an improved global model. Through repeated rounds of local training and centralized aggregation, FL enables collaborative learning while ensuring that sensitive genomic data never leaves the institution in which it was generated. This decentralized training paradigm significantly reduces privacy risks while preserving many of the predictive benefits associated with centralized machine learning.

Recent proof-of-concept studies have demonstrated the feasibility of applying Federated Learning to genomic variant interpretation and have reported predictive performance comparable to conventional centralized training under simulated multi-site environments [7] [8]. These encouraging findings indicate that FL has the potential to overcome one of the most significant barriers preventing large-scale collaborative genomic research. However, several important challenges remain before FL can be effectively deployed in real-world clinical genomics. One of the primary challenges is statistical heterogeneity, commonly referred to as non-Independent and Identically Distributed (non-IID) data, where participating institutions possess datasets with substantially different variant distributions, population characteristics, sequencing protocols and disease prevalence. Such heterogeneity can significantly degrade the convergence and predictive performance of conventional federated optimization algorithms, making robust aggregation strategies an essential component of practical federated learning systems.

In this work, we build upon the promise of Federated Learning by proposing a comprehensive, open-source, full-stack federated learning framework specifically designed for Single Nucleotide Variant (SNV) pathogenicity prediction. Our framework leverages the feature-rich representation introduced by MAGPIE [4] while exploring the effectiveness of Deep Neural Networks (DNNs) as predictive models within the federated setting. Furthermore, we perform a systematic evaluation of multiple advanced federated aggregation strategies that are specifically designed to address the statistical heterogeneity characteristic of real-world multi-institutional genomic datasets. Through this investigation, we aim to establish a practical, scalable and privacy-preserving framework that enables secure collaboration across healthcare institutions without compromising patient confidentiality. Ultimately, this work seeks to provide a robust blueprint for collaborative genomic machine learning that facilitates improved interpretation of Variants of Uncertain Significance (VUS), enhances the accuracy of pathogenicity prediction and contributes toward more equitable and effective genomic diagnostics across diverse patient populations.

Chapter – 2

Literature Review

The interpretation of genomic variants has become one of the most important challenges in modern precision medicine. Advances in next-generation sequencing technologies have dramatically reduced the cost and time required to sequence an individual's genome, enabling Whole Genome Sequencing (WGS) and Whole Exome Sequencing (WES) to become routine diagnostic tools for rare genetic disorders. However, the increased availability of sequencing data has shifted the primary challenge from data generation to data interpretation. A typical human genome contains approximately 4 to 5 million variants relative to the reference genome [2], while only one or a small number of these variants may be responsible for a patient's clinical phenotype. Identifying these disease-causing variants among millions of benign polymorphisms therefore represents a major computational and clinical challenge.

To standardize clinical interpretation, the American College of Medical Genetics and Genomics (ACMG), together with the Association for Molecular Pathology (AMP), proposed a widely adopted framework for variant classification [9]. According to these guidelines, genetic variants are categorized into five classes: Pathogenic, Likely Pathogenic, Variant of Uncertain Significance (VUS), Likely Benign and Benign. This standardized classification enables consistent reporting across clinical laboratories and facilitates evidence-based decision-making during diagnosis. Despite these guidelines, a substantial proportion of observed variants cannot be conclusively classified because of insufficient or conflicting evidence regarding their biological impact. These Variants of Uncertain Significance (VUS) represent one of the most significant challenges in clinical genomics. In several large-scale sequencing studies, VUS account for nearly 60% of reported variants, leaving clinicians unable to determine whether these variants contribute to disease and consequently limiting the diagnostic utility of genomic sequencing.

To address this challenge, numerous computational methods have been developed to predict the pathogenicity of genetic variants. These methods aim to prioritize candidate variants for further clinical investigation by estimating the probability that a particular variant disrupts normal biological function. Over the past two decades, pathogenicity prediction techniques have evolved substantially, progressing from relatively simple conservation-based scoring systems to sophisticated machine learning models capable of integrating diverse sources of genomic and functional information.

The earliest generation of pathogenicity predictors relied primarily on evolutionary conservation and protein sequence information. Among the pioneering approaches, SIFT (Sorting Intolerant From Tolerant) [10] predicts whether an amino acid substitution is likely to affect protein function by measuring the degree of evolutionary conservation across homologous protein sequences. The underlying assumption is that highly conserved amino acid positions are functionally important and substitutions occurring at these positions are more likely to be deleterious. Similarly, PolyPhen-2 (Polymorphism Phenotyping v2) [11] combines sequence conservation with structural and physicochemical properties of proteins to estimate the functional impact of missense variants. These tools represented significant milestones in computational variant interpretation and established the foundation for subsequent machine learning approaches. Nevertheless, their predictive accuracy remained limited because they relied on a relatively small number of biological features and were unable to capture the complex interactions that often determine variant pathogenicity.

As larger annotated variant databases became available, researchers began adopting machine learning techniques capable of integrating multiple sources of biological evidence simultaneously. Rather than relying exclusively on conservation or structural information, these methods combine functional annotations, population allele frequencies, gene-level constraints, regulatory information and other genomic features to improve predictive accuracy.

Among the earliest machine learning approaches, CADD (Combined Annotation Dependent Depletion) [12] represented a major advancement in computational variant scoring. Unlike traditional pathogenicity predictors, CADD estimates the relative deleteriousness of genetic variants by comparing naturally occurring human variants with computationally simulated variants. Instead of directly classifying variants as pathogenic or benign, CADD assigns a continuous score reflecting the likelihood that a variant is evolutionarily deleterious. Although this scoring framework proved highly effective for prioritizing candidate variants, its optimization objective differs from direct clinical pathogenicity classification, limiting its effectiveness in certain diagnostic settings. The original implementation of CADD employed a linear Support Vector Machine (SVM) to integrate genomic annotations and generate deleteriousness scores.

Building upon the CADD framework, DANN (Deleterious Annotation of Genetic Variants using Neural Networks) [5] replaced the linear Support Vector Machine with a Deep Neural Network (DNN). This architectural improvement enabled the model to learn complex nonlinear relationships among genomic features that could not be effectively captured using linear decision boundaries. As a result, DANN achieved a 19% relative reduction in prediction error together with a 14% improvement in Area Under the Curve (AUC) compared with the original CADD model. The success of DANN demonstrated the potential of deep learning techniques for genomic variant interpretation and motivated the adoption of increasingly sophisticated neural network architectures in subsequent pathogenicity prediction models.

An alternative direction in pathogenicity prediction has been the development of ensemble or meta-predictors. Rather than generating genomic features independently, these methods combine the outputs of multiple existing prediction tools into a single integrated model. By leveraging the complementary strengths of several predictors, ensemble models often achieve higher predictive performance than any individual constituent method.

REVEL (Rare Exome Variant Ensemble Learner) [13] is among the most successful examples of this approach. REVEL combines prediction scores generated by thirteen established pathogenicity predictors, including SIFT, PolyPhen-2 and CADD, using a Random Forest classifier. The model is trained using clinically annotated pathogenic and benign variants obtained from ClinVar [3], enabling it to directly optimize clinical pathogenicity prediction rather than general functional impact. REVEL has consistently demonstrated superior performance for missense variant classification and remains one of the most widely adopted ensemble predictors in clinical genomics.

Similarly, ClinPred [14] employs a supervised machine learning framework trained on pathogenic and benign variants while incorporating pathogenicity scores generated by the M-CAP predictor as additional predictive features. Through the integration of multiple complementary sources of information, ClinPred achieves strong discriminative performance in distinguishing disease-associated variants from benign polymorphisms, further demonstrating the effectiveness of ensemble learning approaches in variant interpretation.

Recent advances in machine learning have shifted toward Gradient Boosted Decision Trees (GBDTs), which are particularly effective for handling heterogeneous, high-dimensional feature spaces containing complex nonlinear interactions. Among these methods, MAGPIE (Multimodal Annotation Generated Pathogenic Impact Evaluator) [4] represents the current state of the art in feature-based SNV pathogenicity prediction. MAGPIE is trained using more than 268,000 clinically annotated variants collected from ClinVar and gnomAD [15]. Its predictive strength arises from the combination of the highly efficient LightGBM framework with an extensive feature set generated using ANNOVAR [16]. These features comprise 46 diverse genomic annotations, including evolutionary conservation scores, population allele frequencies, gene-level constraint metrics, functional annotations and outputs from existing pathogenicity predictors. By integrating this rich collection of complementary biological information, MAGPIE achieves an exceptional ROC-AUC of 0.995 for SNV pathogenicity prediction, establishing it as one of the highest-performing centralized models currently available.

Despite its remarkable predictive performance, MAGPIE shares a limitation common to virtually all modern supervised pathogenicity predictors. The model assumes that training data from multiple institutions can be aggregated into a centralized repository before model development. Increasingly stringent privacy regulations governing genomic data sharing make this assumption difficult to satisfy in practical healthcare environments, thereby limiting opportunities for collaborative model development using globally distributed genomic datasets.

Table 2.1: Comparison of Variant Pathogenicity Predictors

Predictor Name	**Core ML Model**	**Key Data Sources**	**Performance (AUC)**
MAGPIE[4]	LightGBM	ClinVar, gnomAD	0.995 (on MAGPIE)
DANN[5]	Deep Neural Network	Observed human vs. simulated variants	0.946 (on ClinVar/ESP)
REVEL[13]	Ensemble of 13 RF	HGMD, Exome Sequencing Project	0.908 (on training set)
ClinPred[14]	Ensemble of RF and GBT	ClinVar, gnomAD	0.9 (on test set)

While feature-based machine learning models have achieved impressive predictive performance, another emerging direction focuses on directly learning biological representations from protein sequences and structures using deep learning. AlphaMissense [17] represents one of the most significant advances in this area. Built upon the protein language modeling capabilities developed for AlphaFold [18], AlphaMissense leverages large-scale protein sequence representations together with evolutionary information obtained from human and primate genomes. Rather than relying primarily on manually engineered genomic annotations, the model learns rich contextual representations that capture structural and functional properties of proteins.

AlphaMissense produces a continuous pathogenicity score between 0 and 1, representing the predicted probability that a missense variant is pathogenic. Based on this score, the model classifies approximately 89% of all possible missense variants, identifying roughly 32% as likely pathogenic and 57% as likely benign while leaving the remaining variants as uncertain. This demonstrates the considerable potential of protein language models for variant interpretation. Moreover, AlphaMissense represents an orthogonal methodology to feature-engineered models such as MAGPIE, suggesting that future pathogenicity prediction systems may benefit from integrating learned protein representations with comprehensive genomic annotations to further improve predictive performance.

Chapter – 3

Problem Definition

The accurate prediction of Single Nucleotide Variant (SNV) pathogenicity is fundamental to the diagnosis of rare genetic diseases. However, the development of highly accurate computational predictors is constrained by a fundamental challenge: while machine learning models require large, diverse, and representative datasets to achieve robust performance, the sensitive nature of genomic data and the legal and ethical requirements governing its use prohibit unrestricted data sharing across institutions. Consequently, valuable genomic datasets remain isolated within institutional silos, limiting model generalizability, introducing population-specific biases, and hindering the interpretation of clinically significant Variants of Uncertain Significance (VUS).

This work addresses this challenge by investigating Federated Learning (FL) as a privacy-preserving alternative to conventional centralized machine learning. The primary objective is to design, implement, and validate a federated learning framework capable of collaboratively training an SNV pathogenicity prediction model across multiple decentralized institutions while ensuring that raw genomic data never leaves its source. By enabling secure collaborative learning, the proposed framework aims to improve predictive performance under realistic distributed settings, mitigate the effects of data fragmentation, and provide a scalable solution for multi-institutional genomic research without compromising patient privacy.

Chapter – 4

Methodology

4.1 Data Preparation

For this project, we leverage the MAGPIE dataset [4], a resource specifically engineered for training variant pathogenecity predictors. The MAGPIE dataset was constructed from the ClinVar dataset by filtering for ”Pathogenic”, ”Likely Pathogenic”, ”Benign” and ”Likely Benign” variants. The ”Pathogenic” and ”Likely Pathogenic” variants make up the positive class for the dataset. Variants from gnomAD with an allele frequency greater than 0.1% were used as the ”Benign” and ”Likely Benign” negative class. The initial set was meticulously filtered to focus on Single Nucleotide Variants (SNVs). The data set also uses ANNOVAR (Annotate Variation) to generate 46 distinct numerical features for each variant. To improve the model performance in classifying rare variants, adding rare benign variants to the training dataset was considered as a trade-off between preserving the importance of allele frequency information and helping identify rare pathogenic variants from rare benign variants.

To evaluate the generalization capability of the proposed model beyond the data used for training, an orthogonal validation dataset was constructed using high-confidence variant annotations obtained from the manually curated SwissProt database. Unlike the training and test datasets, which were derived from the MAGPIE dataset, the orthogonal validation set provides an independent benchmark for assessing the model's ability to accurately classify variants originating from a different data source. Such an evaluation is essential to determine whether the learned model captures generalized biological patterns rather than dataset-specific characteristics.

The construction of the orthogonal dataset involved a series of stringent filtering steps to ensure the reliability and biological relevance of the selected variants. Initially, only exonic Single Nucleotide Variants (SNVs) were retained, ensuring that every selected variant directly affects the coding sequence of a protein and therefore has the potential to alter protein function. Subsequently, synonymous variants were removed, as these mutations do not alter the encoded amino acid sequence and generally have limited impact on protein structure or function. Variants classified as Variants of Uncertain Significance (VUS) were also excluded because their pathogenicity remains unresolved, making them unsuitable as reliable ground-truth labels for model evaluation.

The resulting orthogonal validation dataset therefore consists exclusively of rare, non-synonymous SNVs with well-established functional or clinical annotations. Evaluating the proposed framework on such a dataset provides a stringent assessment of its predictive capability, as these variants are substantially less likely to have been encountered during model training. Consequently, successful performance on the orthogonal dataset demonstrates that the model has learned biologically meaningful relationships between genomic features and pathogenicity rather than merely memorizing patterns present in the training data. This evaluation therefore serves as an important indicator of the framework's robustness, generalizability and potential applicability in real-world clinical genomic analysis, where previously unseen and rare variants are routinely encountered.

To rigorously evaluate the robustness of the proposed federated learning framework under realistic deployment conditions, the MAGPIE dataset was partitioned into multiple simulated Non-Independent

Table 4.1: Summary of Final MAGPIE Training and Validation Datasets

Dataset Name	Total Variants	Pathogenic	Benign
Training Set (ClinVar)	78,018	39,893	38,125
Independent Test Set (ClinVarTest)	8,666	4,356	4,310
Orthogonal Validation Set (SwissProt)	13,383	1,308	12,075

and Identically Distributed (Non-IID) client datasets. Each partition served as the local dataset for an individual federated client, thereby emulating the decentralized data ownership observed in real-world collaborative healthcare environments. Unlike conventional machine learning, where training data are assumed to be independently and identically distributed, federated learning must operate effectively when participating institutions possess datasets with substantially different statistical characteristics.

Constructing a Non-IID data distribution is therefore a critical component in the evaluation of federated learning algorithms. In clinical genomics, data collected across hospitals, diagnostic laboratories and research institutions naturally exhibit considerable heterogeneity arising from differences in patient demographics, disease prevalence, sequencing technologies, variant frequencies and annotation completeness. Consequently, each participating institution may possess a unique local data distribution that differs significantly from that of other institutions. Such statistical heterogeneity poses a substantial challenge to federated optimization, often causing local models to converge toward conflicting objectives and reducing the effectiveness of conventional aggregation algorithms.

By simulating these heterogeneous client datasets, the proposed experimental setup provides a realistic approximation of practical multi-institutional genomic collaborations. This Non-IID partitioning rigorously evaluates the ability of the GENESIS framework to learn a robust global model despite variations in class distributions, population-specific variant frequencies, mutation densities and incomplete genomic annotations across clients. Successfully maintaining high predictive performance under these challenging conditions demonstrates the framework's ability to generalize across diverse clinical environments and validates its suitability for deployment in privacy-preserving collaborative genomic research.

4.2 Feature Engineering

The feature engineering strategy employed in this study transforms raw genomic variant data into a rich set of machine learning features to enhance pathogenicity prediction. The pipeline systematically derives sequence-based features such as allele length, GC content, nucleotide frequencies and transition/transversion status. It aggregates evolutionary conservation scores like phylop and phastcons into summary statistics and computes functional impact features by summarizing variant effect scores and identifying high- and low-impact mutations. Population frequency features are generated by calculating allele frequency statistics and classifying variants as rare or common. Additionally, we construct interaction features between conservation, impact and frequency metrics. Categorical variables such as chromosome

Algorithm 1 Dirichlet-Based Non-IID Dataset Partitioning

Input: Global dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, number of clients K, Dirichlet concentration parameter α, label set $\mathcal{C} = \{0, 1\}$ (Benign, Pathogenic)

Output: Client-local datasets $\{\mathcal{D}_k\}_{k=1}^{K}$

Initialize $\mathcal{D}_k \leftarrow \emptyset$ for all $k \in \{1, \dots, K\}$
for each class $c \in \mathcal{C}$ **do**
    $\mathcal{D}^c \leftarrow \{(x_i, y_i) \in \mathcal{D} \mid y_i = c\}$
    Randomly shuffle $\mathcal{D}^c$
    Sample proportions $\mathbf{p}^c = (p_1^c, \dots, p_K^c) \sim \text{Dir}(\alpha)$
    Compute client allocation sizes: $n_k^c = \lfloor p_k^c \cdot |\mathcal{D}^c| \rfloor$ for $k = 1, \dots, K-1$
    $n_K^c \leftarrow |\mathcal{D}^c| - \sum_{k=1}^{K-1} n_k^c$ ▷ Assign remainder to last client
    Partition $\mathcal{D}^c$ into K disjoint subsets $S_1^c, \dots, S_K^c$ of sizes $n_1^c, \dots, n_K^c$
    for each client $k \in \{1, \dots, K\}$ **do**
        $\mathcal{D}_k \leftarrow \mathcal{D}_k \cup S_k^c$
    end for
end for
return $\{\mathcal{D}_k\}_{k=1}^{K}$

and gene identifiers are label-encoded, with optional support for gene embedding. Reference and alternate alleles are one-hot encoded and the framework supports advanced k-mer and Word2Vec-based sequence embeddings. The pipeline also includes optional feature selection, dimensionality reduction with PCA and feature scaling, resulting in a comprehensive and informative feature set for downstream modeling.

4.3 Model Training

The modeling approach adopts advanced neural network architectures that are specifically tailored for deployment within a federated learning framework [6] [19], enabling collaborative training across multiple decentralized data sources while preserving data privacy. At each participating client site, local data is first preprocessed to ensure consistency and quality before being used to train a primary model composed of several key components. The core architecture integrates a feature extractor with an optional gene embedding layer to better represent complex genomic relationships and incorporates a self-attention mechanism designed to capture the relative importance of different features during the learning process. To support stable and robust training, the model also includes residual blocks and feedforward layers equipped with batch normalization and dropout, which help mitigate overfitting and improve generalization across heterogeneous datasets. In addition to the full architecture, a lightweight model variant without the self-attention module is supported for scenarios with limited computational resources, allowing flexibility in deployment. Furthermore, an ensemble model configuration can be used to aggregate predictions from multiple locally trained models, providing an additional layer of performance enhancement by leveraging diverse learned representations across client nodes. This combination of modular design and federated compatibility makes the approach adaptable to a wide range of environments and data conditions while maintaining strong predictive capabilities.

Model training is carried out locally at each participating client site using both standard training procedures and k-fold cross-validation to ensure robustness and reduce variance in performance estimates. The local training loop consists of per-batch optimization, with optional gradient clipping to stabilize learning and prevent exploding gradients, as well as mechanisms to address class imbalance through weighted random sampling, which helps ensure that pathogenic variants are adequately represented during training. Early stopping criteria based on validation loss are also applied to avoid overfitting and to identify the most effective training point for each client. Once the local training process is complete, model updates—typically in the form of gradients or model weight differences—are securely transmitted to a central server, following federated learning principles that prevent any transfer of raw data. The central server aggregates these updates to refine the global model while maintaining strict data privacy. Throughout the entire training process, performance is continuously monitored using key evaluation metrics such as loss, accuracy, ROC-AUC, precision, recall and F1-score and the best local checkpoints are saved for further analysis. To optimize global model convergence and performance, multiple aggregation strategies are explored, including FedAvg [6], FedProx[20] and SCAFFOLD [21], with all training metrics systematically compared across these approaches. Ultimately, the strategy that yields the strongest performance across distributed datasets will be selected as the final training configuration. This federated training workflow not only ensures rigorous model selection and reliable generalization to unseen data across heterogeneous cohorts, but also preserves privacy by keeping all sensitive information within its original data source.

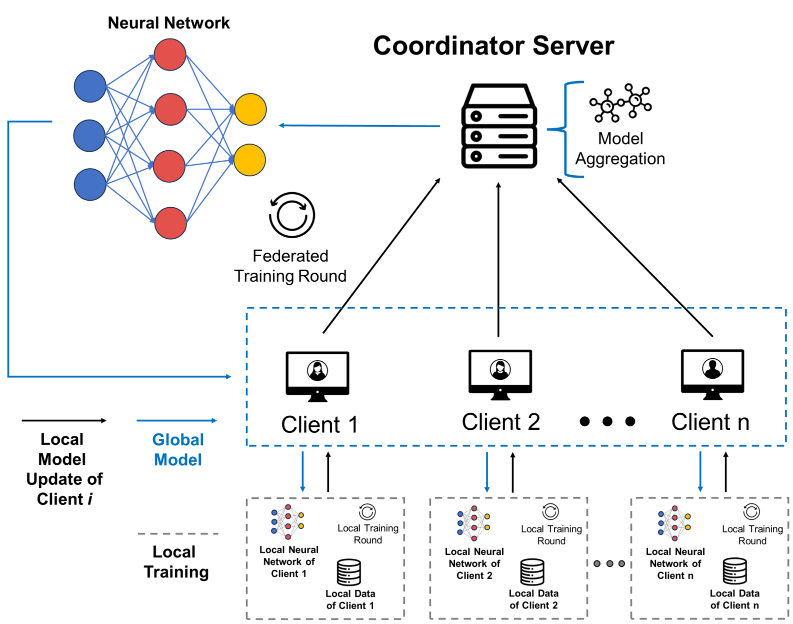


Figure 4.1: FL Architecture

4.4 Model Evaluation

A comprehensive evaluation strategy is employed to rigorously assess the predictive performance of the proposed model and facilitate meaningful comparisons with existing pathogenicity prediction methods. The evaluation is conducted on two distinct datasets: a balanced test set, which enables unbiased assessment of classification performance and an imbalanced orthogonal dataset that more closely reflects the class distribution encountered in real-world clinical scenarios.

To provide a holistic evaluation of model performance, multiple classification metrics are reported. These include accuracy, precision, recall (sensitivity) and the F1-score, each capturing different aspects of predictive capability. Accuracy measures the overall proportion of correctly classified variants, while precision quantifies the proportion of predicted pathogenic variants that are truly pathogenic. Recall evaluates the model's ability to correctly identify pathogenic variants, an especially important consideration in clinical applications where failing to detect a disease-causing variant can have serious consequences. The F1-score, which represents the harmonic mean of precision and recall, provides a balanced measure of performance when both false positives and false negatives are of concern.

Since pathogenic and benign variants are inherently imbalanced in clinical genomic datasets, relying solely on threshold-dependent metrics may provide an incomplete assessment of model performance. Therefore, threshold-independent evaluation is also performed using the Receiver Operating Characteristic (ROC) curve and the corresponding Area Under the Receiver Operating Characteristic Curve (ROC-AUC). The ROC curve illustrates the trade-off between the true positive rate and the false positive rate across different classification thresholds, while the ROC-AUC summarizes the model's overall discriminative ability regardless of the chosen decision threshold. A higher ROC-AUC indicates a greater ability of the model to distinguish pathogenic variants from benign ones.

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$recall = \frac{TP}{TP + FN} \tag{2}$$

$$precision = \frac{TP}{TP + FP} \tag{3}$$

$$F1 - score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{4}$$

$$AUC = \frac{\sum_{i=1}^{n} Rank_i - \frac{n(n+1)}{2}}{n_{positive} * n_{negative}} \tag{5}$$

Collectively, these evaluation metrics provide a comprehensive and robust assessment of the proposed model under both balanced and clinically realistic class distributions. This multi-faceted evaluation framework enables fair comparison with existing pathogenicity prediction models while accounting for the challenges posed by class imbalance and varying classification thresholds.

4.5 Application Development

GENESIS has been designed as a secure, distributed application framework enabling multiple clients to collaboratively participate in a federated learning workflow while maintaining strict privacy and operational control. The system is composed of three major application layers: a Desktop Client Application, a Federated Aggregator Server and a Middleware Orchestrator, all integrated through secure, certificate-based communication channels. This architecture enables seamless coordination across several independent nodes while ensuring that sensitive genetic data remains fully local to each client.

The Desktop Client Application is developed using Electron, providing a cross-platform, native-like interface for users to manage datasets, monitor logs and metrics and run inferences. Electron is paired with LocalStorage for handling lightweight client-side metadata, while sensitive assets such as certificates are stored securely using Keytar, preventing access by other processes. Every client installation comes bundled with a predefined local training script written in Python. Although the training process itself is triggered externally through the orchestrator, the application manages the local environment, displays statuses and interacts with backend services. The desktop client also maintains a persistent WebSocket channel with the orchestrator to receive lifecycle commands like start, stop and heartbeat checks, ensuring real-time coordination during training operations.

The Federated Aggregator Server is implemented using Python and the Flower (FLWR) federated framework, selected for its native support for gRPC-based communication and its modular federation server capabilities. The aggregator exposes secure gRPC endpoints that the clients connect to as part of the training cycle. When the orchestrator initiates a new cycle, the aggregator server is launched, establishes its certificate-based identity and waits for client connections. All communication between clients and the aggregator takes place over mutual TLS (mTLS), guaranteeing that only authenticated and authorized nodes can participate. The aggregator server also maintains its own logs, metrics and model artefacts, which are later consumed by the orchestrator for visualization and historical analysis.

The Middleware Orchestrator forms the central management layer of Genesis and is built using the MERN stack (MongoDB, Express.js, React.js and Node.js). This web platform serves as the control center for administrators and researchers, enabling them to view connected clients, review training histories, inspect logs and issue operational commands. Node.js handles communication with both the aggregator and clients, launching the aggregator's Python process when training begins and relaying events to clients using WebSockets. MongoDB stores all non-sensitive system metadata, including training logs, model versions, connection histories, user roles and inference usage statistics. Role-Based Access Control (RBAC) is built directly into the orchestrator, defining permissions for Admin, Client and Researcher roles. This ensures controlled access to different parts of the system and prevents unauthorized manipulation of the training lifecycle or stored information.

Security is a foundational component of the Genesis architecture. The entire system is bound together using a custom Public Key Infrastructure (PKI) that includes a dedicated Certificate Authority (CA). During onboarding, each client receives a unique set of certificates signed by the CA, generated after applying verification policies. These certificates are used for authenticating connections with both the orchestrator and aggregator. The aggregator and orchestrator themselves also hold CA-signed certificates, ensuring mutual validation at every communication step. Using mTLS across all communication channels (gRPC, REST and WebSockets) protects against impersonation, tampering and man-in-the-middle attacks. Sensitive credential material is never stored in plain text; instead, it is encrypted and placed in secure system stores through Keytar on the client side and environment-based vaulting on the server side.

The operational workflow across the system begins with the admin initiating the training session from the orchestrator dashboard. When the admin clicks “Start Training”, a REST API call is sent from the dashboard to the orchestrator backend, which then launches the aggregated training server by invoking the Python-based Flower aggregator function. The orchestrator immediately returns a response indicating “Server Started”, allowing the dashboard to display this status to the admin. Once the server is confirmed running, the dashboard presents an option “Notify Clients”. When the admin clicks this, the orchestrator uses WebSocket broadcast channels to send a training-start signal to all connected Electron clients.

At the client side, the users do not need to take any action—the moment the Electron application is open, it automatically listens for incoming signals. Upon receiving the start-training event, the Electron app retrieves the securely stored certificates from Keytar, validates the server fingerprint and automatically initiates the local Python client script. This script connects to the aggregator through gRPC and begins participating in the federated learning session. The entire process—from receiving the signal to joining the training round—happens seamlessly without the user needing to press any button. Throughout the session, the orchestrator logs progress, monitors client connectivity and updates the admin dashboard in real time. Once the training completes, the orchestrator cleanly terminates the aggregator server, updates system records and broadcasts a final completion signal to all clients.

Together, these application layers create a cohesive, scalable and strongly secured federated system. The combination of Electron for desktop deployment, MERN for orchestration and Python + Flower for federation ensures a smooth integration of user experience, system management and distributed model coordination. This architecture allows multiple institutions to participate in a high-security environment without compromising the privacy of their genetic datasets, while still providing the administrative and research teams with a unified interface for monitoring system activity, tracking history and maintaining operational control.

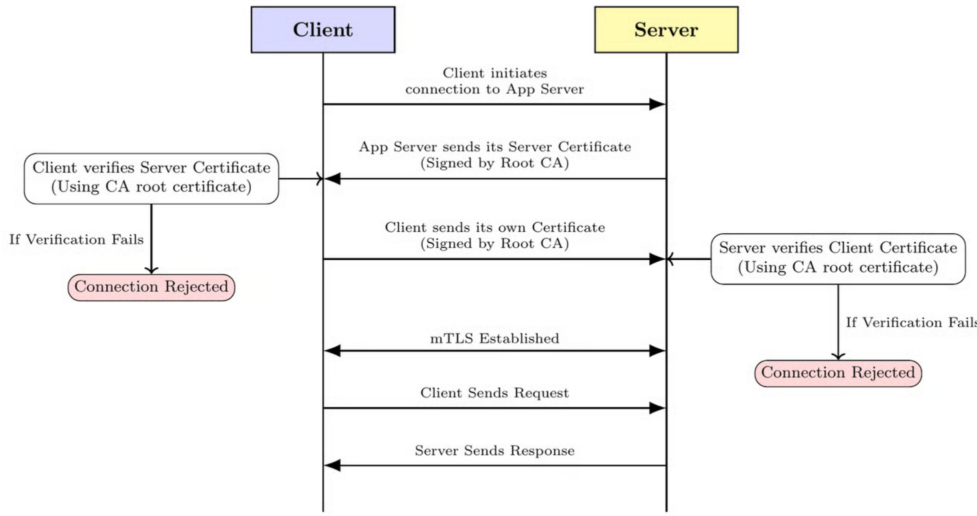


Figure 4.2: mTLS handshake between client and server

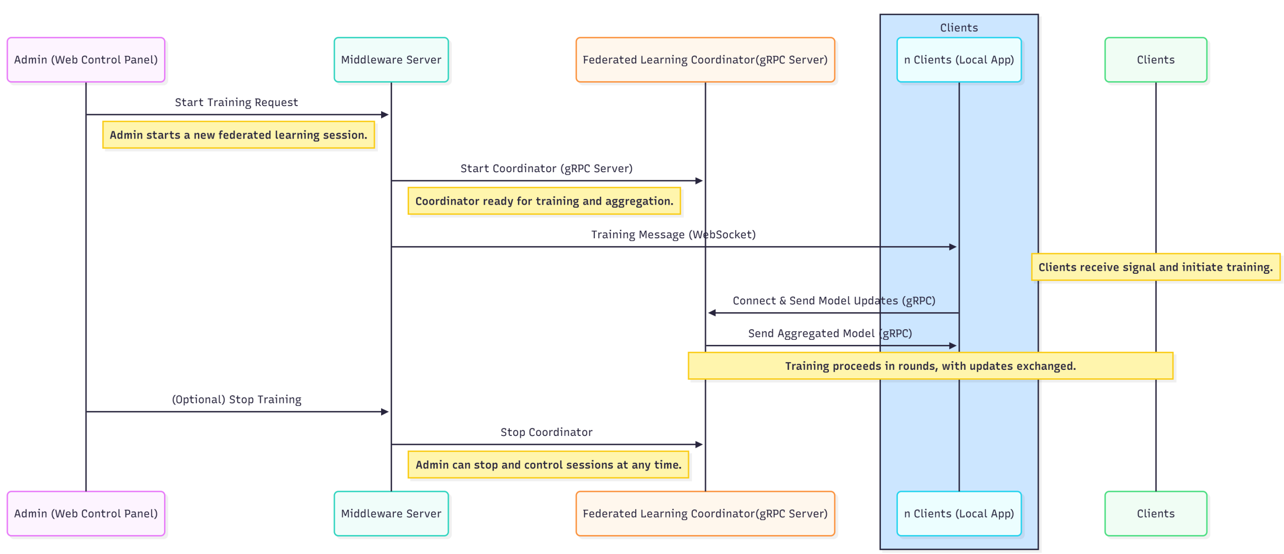


Figure 4.3: Sequence Diagram

Chapter – 5

Facilities Required

5.1 Software Requirements:

Table 5.1: Software Requirements

Category	Technologies / Tools
Datasets	**Clinvar**: Aggregated information about genomic variation and its relationship to human health. **GRCh38**: Largest and most widely used publicly available collection of population variation from harmonized sequencing data.
Data Preparation	**Ensemble VEP**: Reports reference data including gene and variant phenotype associations and population allele frequencies to facilitate variant prioritisation and interpretation. **ANNOVAR**: Utilize update-to-date information to functionally annotate genetic variants detected from diverse genomes.
Federated Framework	**Pytorch**: Open-source machine learning library based on Torch library. **Flower**: Open-source federated learning (FL) framework. **FastAPI**: Open-soruce web framework for building HTTP-based service APIs.
Database	**PostgreSQL**: Open-source object-relational database system. **MongoDB**: open-source NoSQL database. **SQLAlchemy**: Open-source Python library that provides an SQL toolkit and an ORM for database interactions.
Application Development	**Electron**: Open-source software framework to create desktop applications using web technologies. **React.js**: Open-source front-end JavaScript library for building user interfaces. **Node.js**: Open-source, cross-platform JavaScript runtime environment. **Express.js**: Open-source back-end web application framework for Node.js.
Version Control	**GitHub**: Web-based platform that provides version control and collaboration for software development.
Containerization	**Docker**: Set of Paas products that use OS-level virtualization to deliver software in packages.
Deployment	**Koyeb**: High-performance Infrastructure for deploying intensive applications across GPUs, CPUs, and Accelerators.

5.2 Hardware Requirements:

Table 5.2: Hardware Requirements

Category	Specifications
Development Machine	**CPU**: Multi-core processor (Intel i5 / AMD Ryzen 5 5600H or higher). **RAM**: Minimum 16 GB recommended for AI model training. **Storage**: SSD preferred; at least 256 GB free for datasets, models, and logs. **GPU**: NVIDIA GPU with CUDA support for faster model training.
Server / Cloud Instance	**CPU**: Multi-core processors. **RAM**: Minimum 16 GB for AI inference and concurrent requests. **Storage**: Minimum 256 GB for datasets, backups, and logs. **GPU**: GPU or higher for accelerated inference.

Chapter – 6

Experimental Results

The performance of the proposed GENESIS framework was evaluated using a comprehensive set of standard binary classification metrics, including accuracy, precision, recall, F1-score, and the Receiver Operating Characteristic Area Under the Curve (ROC-AUC). These metrics collectively provide a holistic assessment of the model's predictive capability by measuring not only its overall classification accuracy but also its ability to correctly distinguish pathogenic variants from benign variants under varying decision thresholds. Since SNV pathogenicity prediction is a high-impact clinical task, relying on a single evaluation metric is insufficient. Therefore, multiple complementary metrics were considered to ensure a thorough and unbiased evaluation of the proposed framework.

To establish a meaningful benchmark for subsequent federated learning experiments, a baseline model was first trained using the complete, non-partitioned global dataset. This centralized model represents the ideal scenario in which all training data are available at a single location and serves as the upper-performance reference for evaluating the effectiveness of the proposed privacy-preserving framework. The objective of constructing this baseline was to determine whether collaborative training through Federated Learning could produce a model with predictive performance comparable to that of conventional centralized training despite the absence of raw data sharing.

The baseline model was trained for 50 epochs using the architecture shown in Figure 6.1. Throughout training, the model exhibited rapid convergence with minimal evidence of overfitting, indicating that the selected architecture was well suited to the SNV pathogenicity prediction task. Evaluation on the independent test dataset demonstrated excellent generalization capability, achieving a training ROC-AUC of 99.45

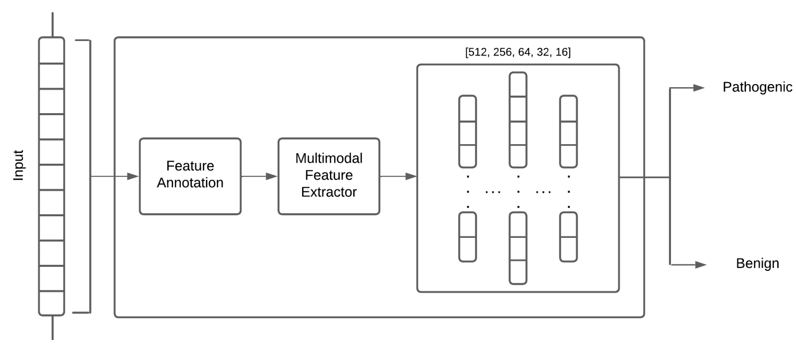


Figure 6.1: Baseline model architecture

Having established the centralized benchmark, all subsequent experiments were conducted within the GENESIS federated learning framework. To ensure that performance comparisons remained fair and scientifically valid, the neural network architecture, hyperparameters, and training configuration of the local models were kept identical to those used by the centralized baseline. Maintaining an identical model architecture ensures that any observed performance differences arise solely from the decentralized learning paradigm and the federated optimization process rather than differences in model capacity or representational power.

To emulate a realistic collaborative clinical environment, the original dataset was partitioned into ten

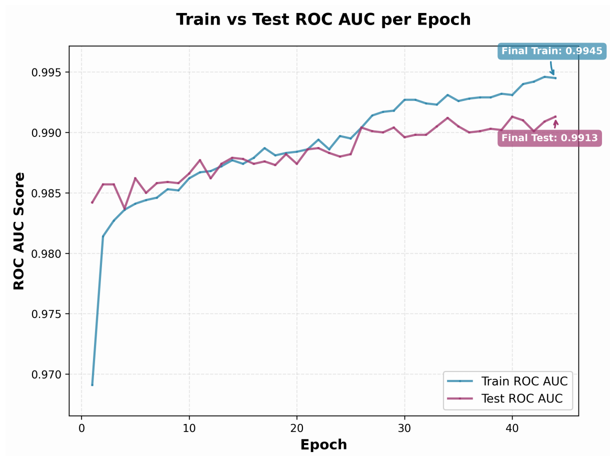


Figure 6.2: Evaluation of the baseline model

statistically heterogeneous (Non-IID) client datasets. Each client independently trained its local model for a maximum of ten local epochs during every federated communication round, with early stopping employed to prevent unnecessary local training once convergence had been achieved. The global model was trained for a total of 75 federated communication rounds. Despite the challenges introduced by decentralized optimization and heterogeneous client distributions, the federated model demonstrated highly competitive predictive performance. The evaluation results indicate that GENESIS successfully preserves much of the predictive capability of the centralized model while simultaneously eliminating the need to exchange sensitive genomic data between participating institutions. Furthermore, the proposed framework achieved performance that was competitive with the state-of-the-art MAGPIE model, demonstrating that privacy-preserving collaborative learning can serve as a practical alternative to conventional centralized training.

Table 6.1: Summary of Evaluation on ROC-AUC

Dataset	**(This Work)**		**MAGPIE [4]**
	GENESIS	**Baseline**	
Training Set	99.84%	99.45%	99.80%
Test Set	99.53%	99.13%	99.50%
Orthogonal Validation Set	96.13%	95.23%	97.00%

Beyond evaluating the overall federated framework, particular emphasis was placed on investigating the impact of different server-side aggregation algorithms on training performance. Since model aggregation is a fundamental component of Federated Learning, the choice of optimization algorithm plays a critical role in determining convergence behavior, robustness to heterogeneous data distributions, and final predictive performance. Three widely used federated optimization strategies were therefore evaluated: FedAvg (Federated Averaging), FedProx (Federated Proximal Optimization), and FedAdam (Federated

Adaptive Optimization).

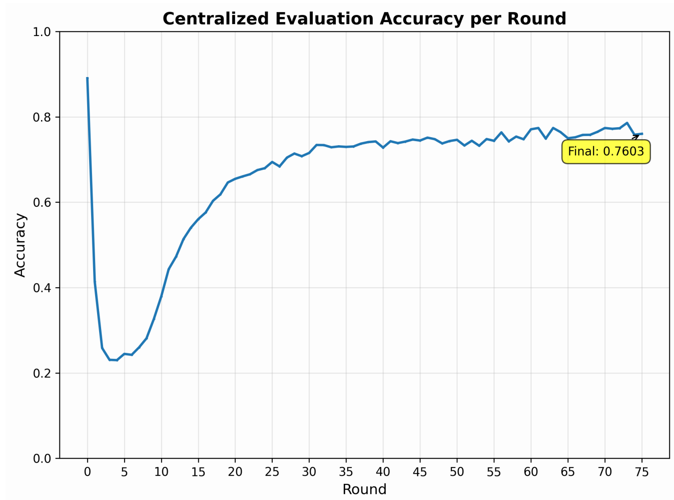
Experimental results clearly demonstrated the limitations of the conventional FedAvg algorithm under heterogeneous client distributions. FedAvg aggregates client model parameters through simple weighted averaging, implicitly assuming that local client objectives are sufficiently similar. However, when client datasets are statistically heterogeneous, as is characteristic of real-world clinical environments, local optimization trajectories tend to diverge substantially. This phenomenon, commonly referred to as *client drift*, causes slower convergence and degrades the quality of the aggregated global model. Consequently, FedAvg consistently produced lower predictive performance than the more advanced optimization algorithms.

Both FedProx and FedAdam substantially improved performance over FedAvg by incorporating mechanisms specifically designed to address the challenges posed by Non-IID data. FedProx introduces a proximal regularization term during local optimization, constraining local model updates to remain closer to the current global model. This additional regularization reduces client drift and produces a significantly smoother optimization trajectory across federated communication rounds. As observed during experimentation, FedProx exhibited highly stable convergence, low variance in evaluation metrics, and consistent improvement throughout training. Although its final predictive performance was marginally lower than that achieved by FedAdam, its stability makes it an attractive choice for practical federated deployments where robustness and reliable convergence are desirable.

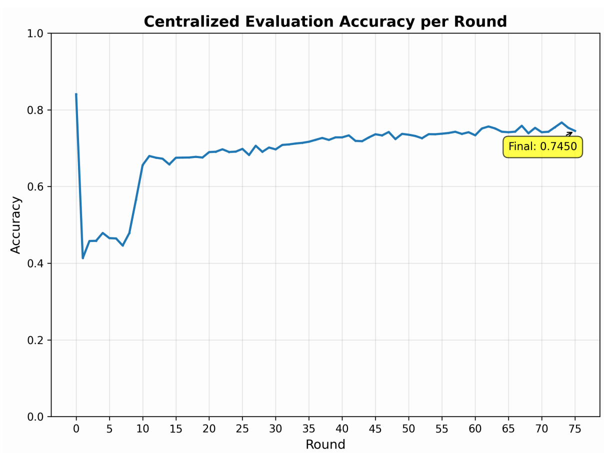
In contrast, FedAdam applies adaptive optimization principles to federated aggregation by incorporating momentum and adaptive learning rate estimation during server-side optimization. This enables faster convergence and allows the global model to achieve superior predictive performance. Experimental evaluation showed that FedAdam consistently produced the highest overall evaluation scores among the three aggregation algorithms. However, this improved predictive performance was accompanied by relatively greater fluctuations during training, indicating a less stable optimization trajectory compared to FedProx. These observations highlight an important trade-off between convergence stability and peak predictive performance when selecting an aggregation strategy for heterogeneous federated learning environments.

Table 6.2: Accuracy of Federated Strategies on Orthogonal Set

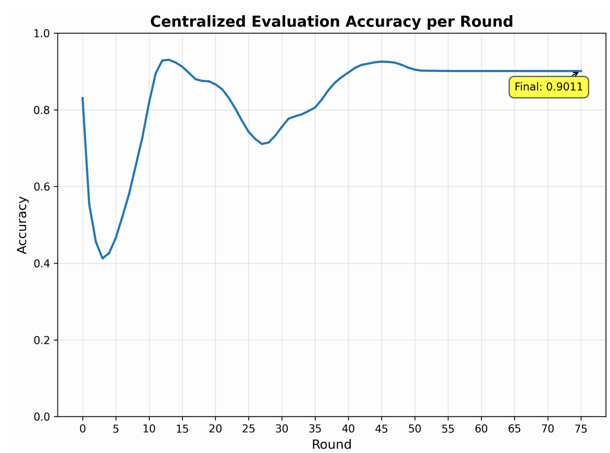
FedAvg	**FedProx**	**FedAdam**
74.50%	76.03%	90.11%



(a) Evaluation of FedAvg



(b) Evaluation of FedProx



(c) Evaluation of FedAdam

Figure 6.3: Comparative evaluation of federated learning strategies

Overall, the experimental results demonstrate that the proposed GENESIS framework is capable of achieving predictive performance comparable to centralized machine learning while preserving complete data privacy through decentralized training. Furthermore, the comparative evaluation of aggregation algorithms confirms that optimization strategies specifically designed for heterogeneous federated environments provide significant advantages over conventional federated averaging. The results therefore validate both the effectiveness of the proposed framework and the importance of selecting aggregation algorithms that explicitly account for statistical heterogeneity in collaborative genomic learning.

Although the current experimental evaluation establishes the feasibility and effectiveness of the proposed framework, additional experimentation remains an important avenue for further improving performance. In particular, a systematic exploration of the hyperparameter space for each aggregation algorithm would provide deeper insight into their optimal operating conditions. For FedProx, varying the proximal regularization coefficient μ could reveal configurations that preserve its observed convergence stability while improving final predictive performance. Similarly, for FedAdam, systematic tuning of server learning rates, momentum coefficients, and adaptive optimization parameters may reduce training instability without sacrificing its superior predictive capability. Such investigations would enable a more comprehensive understanding of the trade-offs between convergence speed, stability, communication efficiency, and predictive accuracy, ultimately identifying the most suitable aggregation strategy for large-scale deployment of the GENESIS framework in real-world collaborative genomic research.

Chapter – 7

Conclusion

This work successfully presents GENESIS, a comprehensive, full-stack, privacy-preserving framework for Single Nucleotide Variant (SNV) pathogenicity prediction based on the Federated Learning (FL) paradigm. The framework was developed to address one of the most significant challenges in modern genomic medicine: reconciling the need for large, diverse datasets required for training high-performing machine learning models with the stringent legal and ethical constraints governing the sharing of sensitive genomic data. By enabling collaborative model training without transferring raw patient data, GENESIS provides a practical solution to the problem of institutional data silos while preserving patient privacy and regulatory compliance.

To emulate the characteristics of real-world collaborative healthcare environments, the available genomic dataset was partitioned into statistically heterogeneous (Non-IID) client datasets. This experimental setup reflects the variations in patient populations, sequencing technologies, disease prevalence and institutional practices that naturally occur across hospitals and research centers. The proposed federated framework demonstrated excellent predictive performance under these challenging conditions, achieving a training ROC-AUC of 99.54% and an independent test ROC-AUC of 99.13%. These results indicate that federated learning can achieve predictive performance comparable to and in some cases exceeding, conventional centralized training while maintaining complete data privacy. The findings therefore validate Federated Learning as a viable and effective approach for large-scale collaborative genomic machine learning.

An important objective of this work was the evaluation of different federated optimization algorithms under heterogeneous data distributions. Experimental results showed that advanced aggregation strategies such as FedProx and FedAdam consistently outperformed the conventional FedAvg algorithm when training on Non-IID datasets. In particular, FedProx demonstrated improved convergence stability and more consistent performance throughout the training process by mitigating client drift caused by heterogeneous local data distributions. These findings highlight the importance of selecting aggregation algorithms that are specifically designed for decentralized learning under realistic clinical conditions, where data heterogeneity is unavoidable.

Beyond its predictive performance, GENESIS establishes a practical framework for secure, collaborative genomic research. The framework incorporates a complete application stack that supports distributed model training through a cross-platform Electron-based desktop application while employing Mutual Transport Layer Security (mTLS) to ensure secure, authenticated and encrypted communication between participating institutions and the central coordinator. By integrating secure communication protocols with a privacy-preserving federated learning pipeline, the framework demonstrates that collaborative genomic analysis can be performed without compromising patient confidentiality or violating data governance regulations.

The broader implications of this work extend beyond the immediate task of SNV pathogenicity prediction. By enabling multiple institutions to collaboratively train machine learning models without exchanging raw genomic data, GENESIS facilitates the development of predictive models trained on larger and more genetically diverse patient populations. Such collaborative learning has the potential to reduce population-specific biases, improve the generalizability of pathogenicity predictors across diverse ancestries and enhance the reliability of computational variant interpretation. Clinically, improved pathogenicity prediction contributes directly to the interpretation of Variants of Uncertain Significance (VUS), which continue to represent one of the most significant bottlenecks in genomic diagnostics. More accurate classification of these variants can shorten diagnostic timelines, assist clinicians in making informed treatment decisions, improve genetic counselling and ultimately enhance healthcare outcomes for patients affected by rare genetic diseases.

The significance of GENESIS lies not only in its implementation but also in the paradigm it represents. Traditional centralized machine learning requires data to be consolidated into a single repository, an approach that is increasingly impractical in the era of stringent privacy regulations. In contrast, GENESIS adopts the principle of bringing the model to the data rather than the data to the model, thereby enabling secure collaboration across institutional boundaries. This shift has the potential to fundamentally transform the way genomic machine learning models are developed, validated and deployed in clinical practice. By combining advanced feature engineering, deep neural network architectures, robust federated optimization techniques and secure software infrastructure within a unified framework, GENESIS provides a scalable foundation for future multi-institutional genomic research.

Although the proposed framework demonstrates the feasibility and effectiveness of federated learning for SNV pathogenicity prediction, several opportunities exist for extending this work. First, the current study evaluates the framework using simulated Non-IID client partitions derived from a centralized dataset. Future work should involve deployment across geographically distributed hospitals, diagnostic laboratories and research institutions to evaluate performance under real-world federated settings with naturally occurring data heterogeneity.

The current implementation primarily focuses on federated optimization while preserving data locality. Additional privacy-enhancing technologies such as Differential Privacy, Secure Multi-Party Computation (SMPC) and Homomorphic Encryption can be incorporated to provide stronger guarantees against information leakage from exchanged model updates. These techniques would further strengthen the framework for deployment in highly regulated clinical environments.

Future research may also investigate more advanced deep learning architectures, including transformer-based genomic models, graph neural networks and multimodal learning approaches capable of integrating genomic, transcriptomic, proteomic and clinical phenotype information. Such multimodal integration has the potential to significantly improve the accuracy and interpretability of pathogenicity prediction, particularly for complex variants whose effects cannot be adequately captured using sequence-based features alone.

Another promising direction is the development of personalized and adaptive federated learning algorithms capable of accounting for institution-specific data distributions while maintaining strong global performance. Personalized federated learning approaches could produce models better suited to local patient populations without sacrificing the benefits of collaborative learning. Additionally, investigating asynchronous federated optimization, client selection strategies, communication-efficient training methods and continual learning would improve the scalability of the framework for large international genomic collaborations.

Finally, the framework can be extended beyond binary SNV pathogenicity prediction to support a broader range of genomic interpretation tasks. These include the prediction of pathogenicity for insertions and deletions (INDELs), copy number variants (CNVs), structural variants (SVs) and the prioritization of candidate disease-causing variants in whole-genome sequencing pipelines. Integration with existing clinical decision-support systems and genomic analysis workflows would further facilitate translation of the proposed framework into routine clinical practice.

In summary, GENESIS demonstrates that privacy-preserving collaborative learning is not only technically feasible but also capable of achieving state-of-the-art predictive performance for genomic variant interpretation. As federated learning technologies continue to mature and healthcare institutions increasingly adopt collaborative artificial intelligence, frameworks such as GENESIS have the potential to become an integral component of next-generation precision medicine, enabling secure, scalable and equitable genomic analysis while preserving the privacy and trust of patients.

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